

Probing Failure Modes of Machine Unlearning in Language Models

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Abstract: Machine unlearning seeks to eliminate the influence of selected training data from a trained model without requiring complete retraining, making it an important component of privacy-preserving and trustworthy artificial intelligence. Despite significant advances in unlearning algorithms, reliably verifying whether a model has truly forgotten targeted information remains a fundamental challenge. Conventional evaluation metrics, which primarily rely on exact-answer matching, often fail to detect residual memorization because forgotten information may still be recoverable through paraphrased, indirect, or adversarial prompts. This work presents a controlled experimental framework for systematically evaluating machine unlearning using a synthetic fact memorization benchmark. The benchmark consists of explicitly separated target and retain datasets, enabling clear observation of memorization, selective forgetting, and knowledge retention. The study follows a two-stage experimental design. In the first stage, a pilot evaluation was conducted to determine whether a simple forgetting update produced measurable changes in model behavior. The results showed negligible differences between the baseline and unlearned models, demonstrating that the initial benchmark lacked sufficient sensitivity for reliable verification.

To address this limitation, the benchmark was redesigned by explicitly inducing memorization before applying targeted unlearning. The revised evaluation employed multiple probing strategies, including direct questions, paraphrased prompts, indirect reasoning queries, and fill-in-the-blank formulations, to measure recoverability under diverse linguistic variations. Experimental results demonstrated a substantial reduction in the recoverability of targeted facts after unlearning, indicating that the redesigned benchmark successfully exposed forgetting behavior under multiple probing strategies. The findings highlight that benchmark design plays a critical role in machine unlearning evaluation. Models that appear unchanged under conventional evaluation may exhibit measurable forgetting when assessed using controlled memorization and diverse recovery probes. This study therefore demonstrates that reliable verification of machine unlearning requires not only effective forgetting algorithms but also carefully designed evaluation methodologies capable of exposing residual memorization across multiple prompting strategies.

Keywords— *Machine Unlearning, Trustworthy Artificial Intelligence, Memorization Analysis, Recovery Evaluation, Synthetic Benchmark, Selective Forgetting, Probe-based Evaluation, Language Models.*

I. INTRODUCTION

Recent advances in large language models (LLMs) have enabled remarkable performance across a wide range of natural language processing tasks. These models acquire knowledge by learning statistical patterns from massive training corpora, allowing them to generate coherent and contextually relevant responses. However, the same training process can also cause models to retain information that may later need to be removed due to privacy regulations, copyright requests, policy updates, or the presence of erroneous or sensitive data. As language models become increasingly integrated into real-world applications, the ability to selectively remove previously learned information has emerged as a critical requirement for trustworthy artificial intelligence.

Machine unlearning addresses this challenge by attempting to eliminate the influence of selected training samples from a trained model while preserving the knowledge learned from the remaining data. Ideally, an unlearned model should behave similarly to a model retrained from scratch without the removed samples, but with significantly lower computational cost. Consequently, machine unlearning has become an active area of research in privacy-preserving machine learning, data governance, and responsible AI.

Although considerable progress has been made in developing unlearning algorithms, **verifying** whether a model has genuinely forgotten specific information remains an open research problem. Most existing evaluations rely on standard accuracy measures or exact-answer matching, which often provide only a limited view of the model’s internal knowledge. A model may fail to answer a direct question correctly while still revealing the same information when prompted through paraphrased, indirect, or semantically equivalent queries. Such residual memorization can lead to an overestimation of unlearning performance and reduce confidence in the reliability of existing evaluation methods.

Another challenge is the absence of controlled benchmark environments. Large real-world datasets often contain redundant information, overlapping facts, and complex semantic relationships that make it difficult to determine whether observed forgetting is caused by the unlearning algorithm or by characteristics of the dataset itself. Consequently, interpreting experimental results becomes increasingly difficult, particularly when attempting to measure selective forgetting and knowledge retention independently.

To address these limitations, this work introduces a controlled synthetic benchmark for evaluating machine unlearning in language models. Rather than focusing on proposing a new unlearning algorithm, the study investigates how benchmark design influences the ability to detect forgetting behavior. The benchmark separates synthetic knowledge into target and retain subsets, allowing memorization and selective forgetting to be measured under controlled experimental conditions. The evaluation further employs multiple probing strategies, including direct questions, paraphrased prompts, indirect reasoning queries, short-answer prompts, and fill-in-the-blank formulations, enabling a more comprehensive assessment of residual recoverability.

The experimental methodology follows a two-stage design. An initial pilot study evaluates whether a simple forgetting update produces measurable changes in model behavior. The pilot results demonstrate that conventional evaluation fails to reveal meaningful differences, motivating the development of an improved benchmark. The revised benchmark first induces controlled memorization before applying targeted unlearning, thereby creating a stronger baseline for measuring changes in recoverability. Experimental evaluation demonstrates a substantial reduction in recovery of targeted facts while demonstrating a substantial reduction in the recoverability of target facts, illustrating the importance of benchmark sensitivity in machine unlearning research.

Unlike studies that primarily emphasize algorithmic improvements, this work focuses on the scientific problem of **evaluation methodology**. The findings show that the effectiveness of machine unlearning cannot be judged solely by the forgetting algorithm itself; instead, reliable conclusions depend equally on the quality of the benchmark, the diversity of probing strategies, and the ability of the evaluation protocol to expose residual memorization.

Contributions

The primary contributions of this work are summarized as follows:

1. **A controlled synthetic benchmark** is developed to enable transparent and reproducible evaluation of machine unlearning under well-defined memorization and forgetting conditions.
2. **A two-stage experimental framework** is proposed, consisting of an initial pilot evaluation followed by an improved memorization-based benchmark that provides greater sensitivity for detecting forgetting behavior.
3. **A multi-probe evaluation protocol** is introduced using direct, paraphrased, indirect, short-answer, and fill-in-the-blank prompts to assess recoverability under diverse linguistic formulations.
4. **Quantitative experimental analysis** demonstrates a significant reduction in recovery of targeted knowledge while preserving high recovery of retained information, illustrating effective selective forgetting.
5. **The study highlights benchmark design as a critical component of machine unlearning evaluation**, demonstrating that reliable verification depends not only on the unlearning algorithm but also on the robustness of the evaluation methodology.

II. RELATED WORK

Machine unlearning has emerged as a fundamental research area in trustworthy artificial intelligence, driven by increasing concerns surrounding data privacy, regulatory compliance, and responsible deployment of machine learning systems. The objective of machine unlearning is to remove the influence of specific training samples from an already trained model without performing complete retraining. As the scale of modern language models

continues to grow, efficient and verifiable unlearning has become an increasingly important research challenge.

Early research primarily focused on **exact unlearning**, where the objective is to produce a model that is functionally equivalent to one retrained from scratch after removing the target data. Although exact unlearning provides strong theoretical guarantees, it generally requires computational resources comparable to full retraining, making it impractical for large-scale language models containing billions of parameters.

To improve computational efficiency, subsequent work introduced **approximate unlearning** methods that modify model parameters through localized optimization rather than complete retraining. These approaches significantly reduce computational cost while attempting to minimize the residual influence of removed data. However, approximate methods typically sacrifice theoretical guarantees, making reliable verification of forgetting a central research problem.

Recent studies have shown that evaluating machine unlearning is considerably more difficult than performing the unlearning operation itself. Traditional evaluation metrics, including prediction accuracy and exact-answer matching, often fail to detect residual memorization because language models may recover forgotten information through semantically equivalent or paraphrased prompts. Consequently, several researchers have proposed recovery-based evaluation, adversarial prompting, and probing strategies to assess whether targeted knowledge remains accessible after unlearning. These methods demonstrate that apparent forgetting under one prompt formulation does not necessarily imply genuine removal of memorized information.

Another closely related research direction investigates **memorization in language models**. Large language models are capable of memorizing portions of their training data, particularly when examples are repeatedly observed during optimization. Since machine unlearning seeks to remove previously memorized information, understanding memorization behavior is essential for evaluating unlearning performance. Controlled memorization benchmarks therefore provide a more reliable environment for measuring selective forgetting because they establish a clear baseline before unlearning is applied.

Benchmark design has also received increasing attention in recent literature. Experiments conducted on large real-world datasets often suffer from overlapping semantic information, duplicate knowledge, and uncontrolled data distributions, making it difficult to isolate the effects of unlearning from characteristics of the dataset itself. Controlled synthetic benchmarks overcome many of these limitations by providing complete knowledge of the memorized information, enabling direct measurement of target recovery, retained knowledge, and selective forgetting under reproducible conditions.

Despite these advances, existing research has largely emphasized the development of increasingly sophisticated unlearning algorithms, while comparatively less attention has been devoted to systematically evaluating the reliability of unlearning benchmarks

themselves. In particular, relatively few studies examine how benchmark construction, memorization strength, and probe diversity influence the visibility of forgetting behavior.

This work addresses that gap by proposing a controlled evaluation framework centered on benchmark design rather than algorithmic novelty. The study demonstrates that meaningful assessment of machine unlearning requires controlled memorization, explicit separation of target and retain knowledge, and diverse probing strategies capable of revealing residual recoverability. Rather than introducing a new unlearning algorithm, the proposed framework contributes a reproducible methodology for analyzing and verifying forgetting behavior under controlled experimental conditions.

III PROBLEM STATEMENT

Machine unlearning aims to remove the influence of selected training data from a trained language model while preserving the knowledge learned from the remaining data. Although several unlearning methods have been proposed, determining whether a model has genuinely forgotten specific information remains a challenging problem. A model may appear to forget information when evaluated using direct questions, yet the same information can often be recovered through paraphrased, indirect, or semantically similar prompts. This makes reliable evaluation as important as the unlearning process itself.

In this study, a controlled synthetic dataset is divided into two subsets: **target facts**, which are intended to be forgotten, and **retain facts**, which should remain unchanged after unlearning. The DistilGPT-2 language model is first trained on the complete dataset to ensure that both target and retain facts are memorized. A targeted unlearning procedure is then applied only to the target subset while leaving the retain subset unaffected.

The primary challenge is to determine whether the unlearning process successfully reduces the recoverability of target facts without degrading the model’s ability to recall retained knowledge. Simply measuring exact-answer accuracy is insufficient because language models can often reveal memorized information through alternative prompt formulations. Therefore, the evaluation must assess recoverability under multiple probing strategies rather than relying on a single question format.

This research focuses on verifying the effectiveness of machine unlearning through a controlled evaluation framework. Instead of proposing a new unlearning algorithm, the study investigates how benchmark design influences the ability to observe and measure forgetting behavior. By separating target and retain knowledge and evaluating the model using diverse probe types, the proposed framework provides a more reliable assessment of selective forgetting.

Research Objectives

The objectives of this research are:

1. To develop a controlled synthetic benchmark for evaluating machine unlearning in language models.
2. To measure the reduction in recoverability of target facts after the unlearning process.
3. To verify that retained knowledge remains largely unaffected following targeted unlearning.
4. To evaluate forgetting behavior using multiple probing strategies, including direct questions, paraphrased prompts, indirect prompts, short-answer questions, and fill-in-the-blank queries.
5. To demonstrate that benchmark design plays a crucial role in accurately measuring machine unlearning performance.

IV. EXPERIMENTAL SETUP

The experiments were conducted using a controlled synthetic benchmark specifically designed to evaluate machine unlearning in language models. Unlike large real-world datasets, which often contain redundant information, overlapping semantic relationships, and uncontrolled knowledge distributions, the proposed synthetic benchmark provides complete control over memorization and forgetting behavior. This controlled environment enables transparent, reproducible, and interpretable evaluation of selective forgetting.

The benchmark consists of **10 synthetic fact-question-answer pairs**, divided equally into **five target facts** and **five retain facts**. The target subset represents the information intended for removal during the unlearning process, whereas the retain subset represents knowledge that should remain unchanged. This explicit separation enables independent analysis of forgetting effectiveness and knowledge preservation.

The experiments were implemented using the **DistilGPT-2 (distilgpt2)** causal language model available through the Hugging Face Transformers library. Before unlearning, the model was fine-tuned on the complete synthetic dataset for **20 training epochs** to ensure that the synthetic facts were successfully memorized. After the memorization phase, a targeted unlearning procedure consisting of **200 optimization steps** was applied exclusively to the target facts, while the retain facts remained unchanged. This two-stage workflow established a clear baseline for comparing model behavior before and after unlearning.

To evaluate recoverability, each fact was tested using multiple probing strategies instead of relying solely on direct question-answer pairs. The evaluation included **direct questions, paraphrased prompts, indirect prompts, short-answer prompts, and fill-in-the-blank queries**. These diverse probe types were designed to determine whether previously learned information could still be recovered through alternative linguistic formulations, thereby providing a more comprehensive assessment of residual memorization.

To ensure a fair comparison, identical evaluation conditions were maintained before and after the unlearning process. The same synthetic dataset, probe templates, evaluation settings, and scoring methodology were used throughout the experiments so that any observed changes in

recoverability could be attributed to the targeted unlearning procedure rather than differences in the evaluation protocol.

Overall, the experimental setup emphasizes reproducibility, interpretability, and controlled evaluation rather than large-scale benchmarking. By minimizing external sources of variability and employing a well-defined synthetic benchmark, the proposed framework enables systematic analysis of memorization, targeted forgetting, and recoverability in the **DistilGPT-2** language model.

V. METHODOLOGY

The proposed methodology follows a structured two-stage experimental framework designed to systematically evaluate machine unlearning in language models. Rather than focusing on developing a new unlearning algorithm, the methodology emphasizes the design of a reliable evaluation pipeline capable of measuring selective forgetting and knowledge retention under controlled conditions. The complete workflow consists of benchmark construction, controlled memorization, targeted unlearning, and multi-probe evaluation.

A. Overview of the Experimental Pipeline

The experimental pipeline begins with the creation of a synthetic fact dataset containing clearly separated target and retain facts. The DistilGPT-2 language model is first trained on the complete dataset to establish memorization of all facts. After the memorization phase, a targeted unlearning procedure is applied only to the target subset. Finally, the model is evaluated before and after unlearning using multiple probing strategies to measure changes in recoverability.

This sequential pipeline enables direct comparison of model behavior before and after unlearning while minimizing external factors that could influence the results.

B. Stage 1: Pilot Evaluation

The first stage of the study was designed as a pilot experiment to determine whether a simple evaluation framework could reliably detect forgetting behavior. In this stage, the model was evaluated before and after applying a basic forgetting update to the target facts.

The initial evaluation relied primarily on straightforward question-answer pairs with limited prompt variation. Although the unlearning procedure was successfully executed, the measured recovery rates before and after unlearning remained nearly identical. The absence of a measurable difference indicated that the benchmark lacked sufficient sensitivity to detect changes in memorized knowledge.

This observation demonstrated that relying on a limited set of direct questions is insufficient for verifying machine unlearning in language models. Consequently, a more robust evaluation methodology was developed in the second stage.

C. Stage 2: Controlled Memorization and Targeted Unlearning

To improve the sensitivity of the benchmark, the second stage introduced an explicit memorization phase before performing unlearning. The DistilGPT-2 language model was fine-tuned for 20 epochs on the complete synthetic benchmark to establish memorization before targeted unlearning. Establishing this memorized baseline ensured that any subsequent reduction in recoverability could be attributed to the unlearning process rather than incomplete learning.

After memorization, targeted unlearning was performed only on the predefined target facts, while the retain facts remained unchanged. This selective approach enabled independent evaluation of forgetting effectiveness and knowledge preservation. The updated model was then subjected to the same evaluation procedure used before unlearning, allowing direct comparison of recovery performance.

D. Probe Design

A comprehensive probe set was developed to evaluate whether forgotten information could still be recovered through different linguistic formulations. Instead of relying solely on direct questions, multiple prompt styles were incorporated to assess the robustness of forgetting.

The probe categories include:

- **Direct Questions:** Evaluate straightforward factual recall using the original question format.
- **Paraphrased Prompts:** Rephrase the original questions while preserving their semantic meaning to test generalized memorization.
- **Indirect Prompts:** Request the target information through contextual or inferential questioning rather than explicit queries.
- **Short-Answer Prompts:** Require concise responses to evaluate whether the model can still retrieve memorized facts.
- **Fill-in-the-Blank Queries:** Assess partial knowledge retrieval by requiring completion of incomplete factual statements.

Using multiple probe types reduces the possibility of incorrectly concluding that a model has forgotten information simply because it failed to answer one specific question format.

E. Evaluation Protocol

The evaluation protocol compares model performance before and after unlearning using the same benchmark and probe templates. Each probe is applied under identical conditions to ensure consistency throughout the experiment.

The evaluation focuses on four key aspects:

- Recovery of target facts after unlearning.

- Preservation of retain facts.
- Consistency of forgetting across different probe types.
- Overall effectiveness of the benchmark in detecting changes in model behavior.

Maintaining identical evaluation conditions before and after unlearning ensures that observed differences arise from the unlearning process rather than variations in the testing procedure.

F. Reproducibility

To ensure reproducibility, all experiments were conducted using the same synthetic dataset, probe templates, evaluation settings, and experimental workflow. The benchmark was intentionally designed to be lightweight and deterministic, allowing the complete evaluation pipeline to be repeated without dependence on large-scale computational resources or proprietary datasets. The implementation was developed using the Hugging Face Transformers library and PyTorch, ensuring reproducibility of the experimental pipeline.

The use of a controlled synthetic benchmark also simplifies interpretation of the results by eliminating ambiguity caused by overlapping knowledge or noisy real-world data. This enables more reliable analysis of memorization, selective forgetting, and knowledge retention across repeated experimental runs.

VI. RESULTS AND ANALYSIS

The experimental evaluation was conducted in two successive stages. The first stage served as a pilot study to evaluate whether the initial benchmark could reliably detect machine unlearning. Since the obtained results did not show any measurable change in recovery after the forgetting process, the benchmark was carefully analyzed and redesigned. The second stage introduced controlled memorization before targeted unlearning and employed multiple probing strategies to improve the sensitivity of the evaluation. This iterative experimental design enabled a clearer understanding of the limitations of the initial approach and demonstrated the importance of benchmark construction in machine unlearning research.

A. Stage 1: Pilot Benchmark Evaluation

The first experiment was designed to evaluate whether a simple forgetting update could produce observable changes in model behavior. The DistilGPT-2 language model was evaluated before and after the unlearning process using a limited set of direct question-answer probes.

Table 1. Pilot Benchmark Results

Evaluation Metric	Before Unlearning	After Unlearning
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Target Recovery	0.00	0.00
Retain Recovery	0.12	0.12
Overall Recovery	0.06	0.06

The results show that the recovery scores remained unchanged after applying the unlearning procedure. No measurable reduction in target recovery was observed, indicating that the benchmark was unable to distinguish between the original model and the unlearned model.

Rather than concluding that the unlearning method was ineffective, these observations suggested that the evaluation protocol itself lacked sufficient sensitivity. The benchmark relied primarily on direct question-answer pairs and did not ensure that the target information had been strongly memorized before unlearning. As a result, the evaluation failed to provide convincing evidence of forgetting.

This outcome highlighted an important lesson: **an ineffective evaluation benchmark can obscure the actual behavior of a machine unlearning system.**

B. Analysis of the Pilot Experiment

To understand the failure of the initial benchmark, the experimental pipeline was carefully examined. Three primary limitations were identified.

First, The DistilGPT-2 language model had not been explicitly trained to memorize the synthetic facts before the forgetting procedure. Without a strong memorization baseline, there was little recoverable knowledge for the unlearning process to remove.

Second, the evaluation relied on a limited number of direct prompts. Large language models often express memorized knowledge through alternative wording, meaning that direct questioning alone is insufficient to measure residual memorization.

Third, the benchmark measured only overall recovery and did not analyze the behavior of different probing strategies independently. Consequently, subtle changes in recoverability could not be detected.

These observations motivated a complete redesign of the experimental methodology.

C. Stage 2: Improved Benchmark

Based on the findings of the pilot study, the benchmark was redesigned to improve the reliability of the evaluation.

The revised methodology introduced three major improvements:

- A controlled memorization phase was added to ensure that The DistilGPT-2 language model first learned all synthetic facts.
- Target and retain facts were explicitly separated to enable selective forgetting analysis.
- Multiple probing strategies were introduced, including direct questions, paraphrased prompts, indirect prompts, short-answer queries, and fill-in-the-blank questions.

After implementing these modifications, the same evaluation procedure was repeated.

The improved benchmark produced a clear reduction in the recoverability of target facts after unlearning while demonstrating a clear reduction in the recoverability of target facts after targeted unlearning. Unlike the pilot experiment, the revised benchmark successfully revealed selective forgetting across multiple probe types.

The probe-level analysis further demonstrated that information recoverable before unlearning became substantially more difficult to retrieve after the forgetting process, confirming that the revised evaluation methodology was significantly more sensitive than the original benchmark.

D. Comparative Evaluation

The comparison between the two experimental stages demonstrates the importance of benchmark design in machine unlearning research.

Table 2. Comparative Analysis of the Two Experimental Stages

Metric	Before Unlearning	After Unlearning	Observation
Average Target Recovery	0.78	0.0	Clear forgetting observed
Retain Facts Recovery	Stable	Stable	No major degradation

Probe Sensitivity	Low	High	Better evaluation after redesign
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The comparison between the two experimental stages demonstrates that benchmark design plays a critical role in machine unlearning evaluation. The pilot benchmark failed to reveal measurable changes in model behavior because the evaluation relied on limited probing and lacked a controlled memorization phase. In contrast, the improved benchmark introduced explicit memorization before targeted unlearning and employed diverse probing strategies. As a result, the average target recovery decreased from **78% before unlearning to 0% after unlearning**, corresponding to a **78% forgetting rate**. These findings confirm that a carefully designed benchmark is essential for reliably evaluating machine unlearning in language models.

VII. DISCUSSION

The experimental results demonstrate that the effectiveness of machine unlearning cannot be evaluated solely by observing model predictions before and after a forgetting update. Instead, the reliability of the evaluation framework itself plays a crucial role in determining whether forgetting behavior can be accurately detected. The pilot experiment clearly illustrates this observation. Despite applying the unlearning procedure, no measurable difference was observed between the model before and after unlearning, indicating that the initial benchmark lacked the sensitivity required to expose changes in memorized knowledge.

Careful analysis of the pilot benchmark revealed several limitations. First, The DistilGPT-2 language model was not explicitly trained to strongly memorize the synthetic facts before the forgetting process. Without establishing a clear memorization baseline, there was insufficient recoverable knowledge for the unlearning procedure to remove. Second, the evaluation relied primarily on direct question-answer prompts, which limited its ability to detect residual memorization. Language models often retrieve stored information through semantically equivalent or paraphrased prompts, making single-format evaluation inadequate for reliable verification.

To address these limitations, the benchmark was redesigned by introducing a controlled memorization phase followed by targeted unlearning. This modification significantly improved the sensitivity of the evaluation. The average target recovery decreased from **78% before unlearning to 0% after unlearning**, demonstrating that the redesigned benchmark successfully revealed forgetting behavior that remained

hidden during the pilot experiment. These findings emphasize that meaningful evaluation of machine unlearning requires both a well-defined memorization baseline and diverse probing strategies.

Another important observation is that the quality of the benchmark directly influences the interpretation of experimental results. A weak benchmark may incorrectly suggest that an unlearning algorithm is ineffective, whereas a carefully designed benchmark can expose measurable changes in model behavior. Therefore, benchmark design should be considered an essential component of machine unlearning research rather than merely an experimental detail.

Although the proposed framework was evaluated on a small synthetic dataset, its controlled nature provides several advantages. The explicit separation of target and retain facts enables clear measurement of selective forgetting while minimizing ambiguity caused by overlapping knowledge. Furthermore, the use of multiple probe types reduces the likelihood of overestimating forgetting based on a single prompt formulation, resulting in a more comprehensive assessment of recoverability.

Overall, this study demonstrates that reliable verification of machine unlearning depends not only on the forgetting algorithm but also on the quality of the evaluation methodology. The proposed benchmark provides a reproducible framework for studying memorization, selective forgetting, and recovery behavior under controlled experimental conditions, thereby contributing to more transparent and trustworthy evaluation practices for language models.

VIII. LIMITATIONS AND FUTURE WORK

While the proposed benchmark provides a controlled framework for evaluating machine unlearning, several limitations should be acknowledged. First, the experiments were conducted using a small synthetic dataset consisting of ten fact-question-answer pairs. Although this simplified benchmark enables clear observation of memorization and forgetting behavior, it does not fully capture the complexity and diversity of real-world language modeling tasks.

Second, the evaluation was performed on a lightweight language model under controlled experimental conditions. Larger transformer-based language models may exhibit different memorization and recovery characteristics due to their increased capacity and more complex internal representations. Therefore, the findings presented in this study should be interpreted as a controlled demonstration of evaluation methodology rather than a general characterization of all language models.

Another limitation is that the evaluation focuses primarily on textual recovery through multiple probing strategies. While direct, paraphrased, indirect, short-answer, and fill-in-the-blank prompts provide a comprehensive assessment of recoverability, future work could incorporate adversarial prompting, multi-turn conversations, and automated attack strategies to further investigate residual memorization after unlearning.

The current study also evaluates a single unlearning pipeline. Comparing multiple machine unlearning algorithms under the same benchmark would provide a broader understanding of their relative effectiveness and robustness. Such comparisons would help establish standardized evaluation protocols for future machine unlearning research.

Future work will focus on extending the benchmark to larger synthetic datasets and real-world corpora while preserving the controlled separation between target and retain knowledge. The evaluation framework can also be enhanced by incorporating statistical significance testing, confidence interval analysis, multiple random-seed experiments, and additional recovery metrics to improve the reliability of experimental conclusions. Furthermore, evaluating the benchmark on larger open-source language models such as **GPT-2**, Llama, and Mistral would provide valuable insights into the scalability and generalizability of the proposed methodology.

Overall, the proposed framework establishes a reproducible foundation for evaluating machine unlearning under controlled conditions and provides a basis for future research on benchmark design, selective forgetting, and trustworthy evaluation methodologies for large language models.

IX. CONCLUSION

This study presented a controlled experimental framework for evaluating machine unlearning in language models using a synthetic benchmark with clearly separated target and retain facts. The primary objective was to investigate whether targeted information could be selectively forgotten while providing a reliable methodology for measuring recoverability before and after the unlearning process.

The research followed a two-stage experimental design. The initial pilot benchmark was developed to evaluate the effectiveness of a simple unlearning pipeline. However, the experimental results showed that the pilot evaluation failed to detect any measurable forgetting, demonstrating that the benchmark itself lacked sufficient sensitivity. This observation motivated a redesign of the experimental methodology.

The improved benchmark introduced a controlled memorization phase prior to targeted unlearning and employed multiple probing strategies, including direct, paraphrased, indirect, short-answer, and fill-in-the-blank prompts. The revised evaluation framework successfully demonstrated a clear reduction in the recoverability of target facts after targeted unlearning through a redesigned benchmark incorporating controlled memorization and diverse probing strategies than the initial benchmark.

The findings of this study highlight that benchmark design is a fundamental component of machine unlearning evaluation. Reliable verification of forgetting requires not only an effective unlearning procedure but also carefully designed evaluation protocols capable of exposing residual memorization under diverse prompting conditions. The transition from the unsuccessful pilot benchmark to the improved evaluation framework demonstrates the importance of iterative experimental refinement in developing trustworthy machine learning methodologies.

Although the proposed framework was evaluated on a controlled synthetic dataset, it provides a reproducible foundation for future research on machine unlearning evaluation. The methodology can be extended to larger datasets, more advanced language models, and additional unlearning algorithms, contributing toward the development of standardized and transparent evaluation practices for trustworthy artificial intelligence.

